

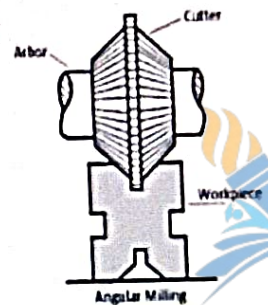
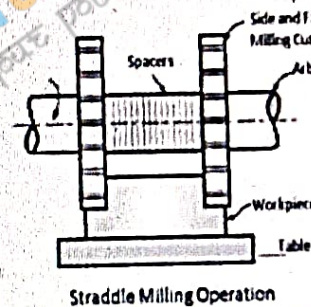
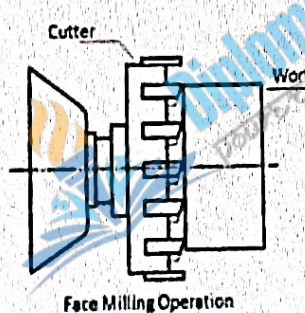
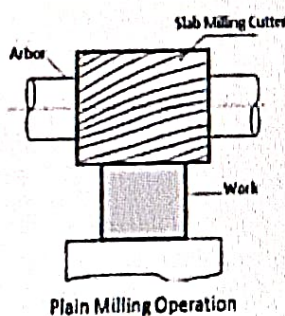
Machine Tool Technology Week-9

1. Explain Plain Milling Operation
2. Prepare the Process plan as per the given drawing (Solid Block) for Plain Milling
3. Estimate the Production cost per Piece as per the given drawing for Plain Milling Operation (Consider all Direct and Indirect costs)

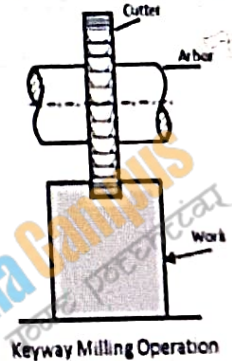
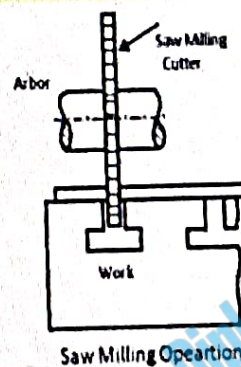
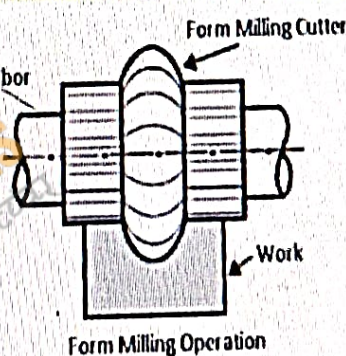
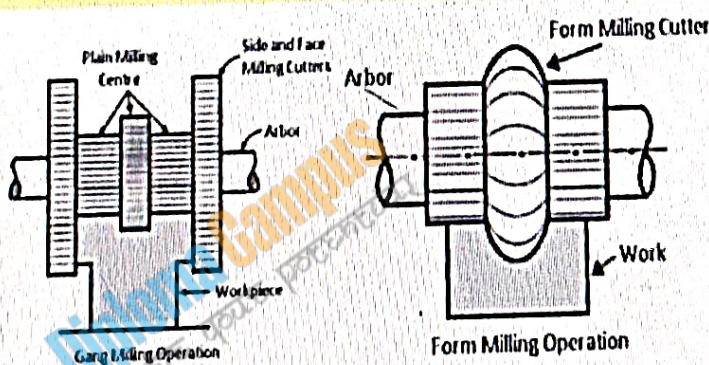
Milling Machine Operations

Following are the different types of operations performed on milling machine:

1. Plain Milling Operation
2. Face Milling Operation
3. Side Milling Operation
4. Straddle Milling Operation
5. Angular Milling Operation
6. Gang Milling Operation
7. Form Milling Operation
8. Profile Milling Operation
9. End Milling Operation
10. Saw Milling Operation
11. Milling Keyways, Grooves, and Slot
12. Gear Milling



Types of Milling Operations



1. Plain Milling Operation

- The plain milling is the most common types of milling machine operations.
- Plain milling is performed to produce a plain, flat, horizontal surface parallel to the axis of rotation of a plain milling cutter.
- The operation is also known as slab milling.
- To perform the operation, the work and the cutter are secured properly on the machine.
- The depth of cut is set by rotating the vertical feed screw of the table. And the machine is started after selecting the right speed and feed.

2. Face Milling Operation

- The face milling is the simplest milling machine operations
- This operation is performed by a face milling cutter rotated about an axis perpendicular to the work surface.
- The operation is carried in plain milling, and the cutter is mounted on a stub arbor to design a flat surface.
- The depth of cut is adjusted by rotating the cross feed screw of the table.

3. Side Milling

- The side milling is the operation of producing a flat vertical surface on the side of a workpiece by using a side milling cutter.
- The depth of cut is set by rotating the vertical feed screw of the table

4. Straddle Milling

- The straddle milling is the operation of producing a flat vertical surface on both sides of a workpiece by using two side milling cutters mounted on the same arbor.
- Distance between the two cutters is adjusted by using suitable spacing collars.
- The straddle milling is commonly used to design a square or hexagonal surfaces.

5. Angular Milling Operation

- The angular milling is the operation of producing an angular surface on a workpiece other than at right angles of the axis of the milling machine spindle
- The angular groove may be single or double angle and may be of varying included angle according to the type and contour of the angular cutter used. It is the production of V-blocks.
- One simple example of angular milling is the

6. Gang Milling

- The gang milling is the operation of machining several surfaces of a workpiece simultaneously by feeding the table against a manner of cutters having the same or different diameters mounted on the arbor of the machine
- The method saves much of machining time and is widely used in repetitive work
- Cutting speed of a gang of cutters is calculated from the cutter of the largest diameter



7. Form Milling Operation

- The form milling is the operation of producing the irregular contour by using form cutters.
- The irregular shape must be convex, concave, or of any other shape. After machining, the formed surface is inspected by a template gauge. Cutting rate for form milling is 20% to 30% less than that of the plain milling.

8. Profile Milling Operation

- The profile milling is the operation of reproduction an outline of a template or complex shape of a master dies on a workpiece.
- Different cutters are used for profile milling. An end mill is one of the widely used milling cutters in profile milling work

9. End Milling Operation

- The End milling is the operation of producing a flat surface which may be vertical horizontal or at an angle in reference to the table surface.
- The cutter used is an end mill. The end milling cutters are also used for the production of slots, grooves or keyways.
- A vertical milling machine is more suitable for end milling operation

10. Saw Milling Operation

- Saw-milling is the operation of producing narrow slots or grooves on a workpiece by using a saw-milling cutter.
- The saw milling also performed for complete parting-off operation.
- The cutter and the workpiece are set in a manner so that the cutter is directly placed over one of the T-slots of the table.

11. Milling Keyways, Grooves, and Slot

- The operation of producing of keyways, grooves and slots of varying shapes and sizes can be performed in a milling machine
- It is done by using a plain milling cutter, a metal slitting saw, an end mill or by a side milling cutter
- The open slots can be cut by a plain milling cutter, a metal slitting saw, or by a side milling cutter. The closed slots are produced by using endmills
- A dovetail slot or T-slot is manufactured by using special types of cutters designed to give the required shape on the workpiece.
- The second slot is cut at right angles to the first slot by feeding the work past the cutter.
- A woodruff key is designed by using a woodruff key slot cutter.
- Standard keyways are cut on the shaft by using side milling cutters or end mills
- The cutter is set exactly at the centre line of the workpiece and then the cut is taken.

12. Gear Milling

- The gear cutting operation is performed in a milling machine by using a form-relieved cutter. The cutter may be a cylindrical type or end mill type
- The cutter profile fits exactly with the tooth space of the gear.



- Equally spaced gear teeth are cut on a gear blank by holding the work on a universal dividing head and then indexing it.

Aim- Produce a rectangular block by face milling and prepare a slot on one face with a slotter cutter/side and face cutter

Materials used- mild steel rectangular block as per drawing

Cutting tool used- slotter cutter, side and face cutter

Theory- A milling machine is one of the most important machine tools. It removes metals from the work piece it feeds against rotating cutting tool used. In a milling machine it has multiple cutting edges which remove metal at a faster rate. Milling machine is a point of application in the production of gears and cutting tools.

TYPES OF MILLING MACHINE

The milling machine is classified as follows:

- 1) Column and knee type
- 2) Horizontal milling machine
 - a) vertical milling machine
 - b) Universal milling machine
- 3) Bed type milling machine
- 4) Planer type

PRINCIPLE OPERATIONS AND PARTS OF A MILLING MACHINE

The principle parts of the column and knee type milling machines are as follows:

BASE

It is the foundation of the machine. It carries the column and gives strength and rigidity to the machine. It serves as the reservoir for cutting fluid.

COLUMN

It is the main supporting frame movement vertically on the base.

KNEE

It projects from the column and slides up and down on the vertical guide ways of the column. It supports the saddle and knee table. The vertical position of the knee can be adjusted by an elevating screw provided at the bottom of the knee.

SADDLE

It supports the table and can be adjusted transversely on the guide ways provided on the top surface of the knee. These guide ways are exactly at 90° to the face of the column.

TABLE

It slides on the guide ways of saddle and travels longitudinally in a horizontal plane. It supports the work piece and other features on it

OVERHANGING ARM

It is mounted on the top of the column in turn extend beyond the column face. ARBOR it may be considered as extension of the spindle.

MILLING OPERATIONS:

The different types of operations that can be performed in milling machine are given below:

Plain milling

Angular milling

Face and side milling

Gang milling

End milling

Straddle milling.

PLAIN MILLING

It is a method of producing flat surface. It is parallel to the axis of the cutter. It is held in the milling arbor.

FACE AND SIDE MILLING

It is a method of producing flat surface right angle to the axis of cutter and in turn in this method large flat surface can be obtained.

GANG MILLING

It is a method of milling by using two or more cutters arrangement together on one arbor. It is used the similar pieces must be made.

ANGULAR MILLING

It is a method of producing a flat surface at an angle to the axis of the cutter. Angular surface are milled by tilting the spindle.

END MILLING

It is a method of milling slots keyway surface by end mills and also it is used in milling cutters.

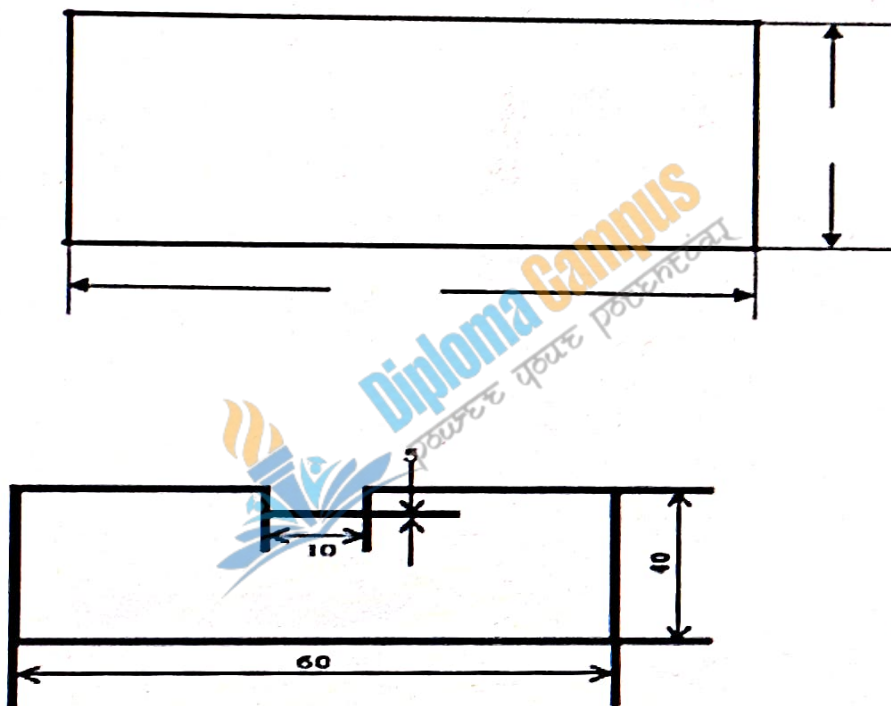
STRADDLE MILLING

It is a method of milling to parallel slides at a work by employing to side milling.

Procedure:-

1. Fix the work piece on the worktable and then fixing milling cutter on the tool holder.
2. Select a suitable cutting speed and feed.

3. Work piece fed to the cutter atomically or manually.
4. The work piece is fed against the direction of the milling cutter.
5. The surface is cleaned with the help of brush to remove the chips.
6. This procedure is followed until the required dimension is obtained



. **Result:** The model is prepared as per the drawing given.



Example 35. Estimate the time required to prepare a square prism on a milling machine from a round block 90 mm. length, 40 mm. in diameter, feed 25 mm./minute and set up and indexing time is 15 minutes.

Solution.

Here, length of cut = 90 mm.

Feed = 25 mm/mic.

Let added table travel

= 10 mm.

$$\therefore T_1 = \frac{90+10}{25} \times 4,$$

where 4 is the sides of square prism i.e. No. of Indexing

$$\therefore T_1 = 16 \text{ minute}$$

Here set up and indexing time,

$$T_2 = 15 \text{ min.}$$

$\therefore$ Total time of milling, T

$$= T_1 + T_2$$

$$= 16 + 15$$

$$= 31 \text{ minute. Ans.}$$

Example 36. Estimate the milling time to cut 60 teeth on a Gear blank 60 mm thick, feed 35 mm/minute and take overall set up time as 10 minutes.

Solution.

$$T = \frac{\text{Length of cut}}{\text{Feed/minute}} \times \text{No. of Index} + \text{set up time.}$$

Here, Set up time = 10 minutes

Length of cut = 60 mm.

Let table travel = 10 mm.

No. of Index = No. of teeth

$$= 60,$$

Feed/minute = 35 mm.

$$\therefore T = \frac{60+10}{35} \times 60 + 10$$

$$= 130 \text{ minutes. Ans.}$$

